Brain Activity in Self- and Value-Related Regions in Response to Online Antismoking Messages Predicts Behavior Change

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A

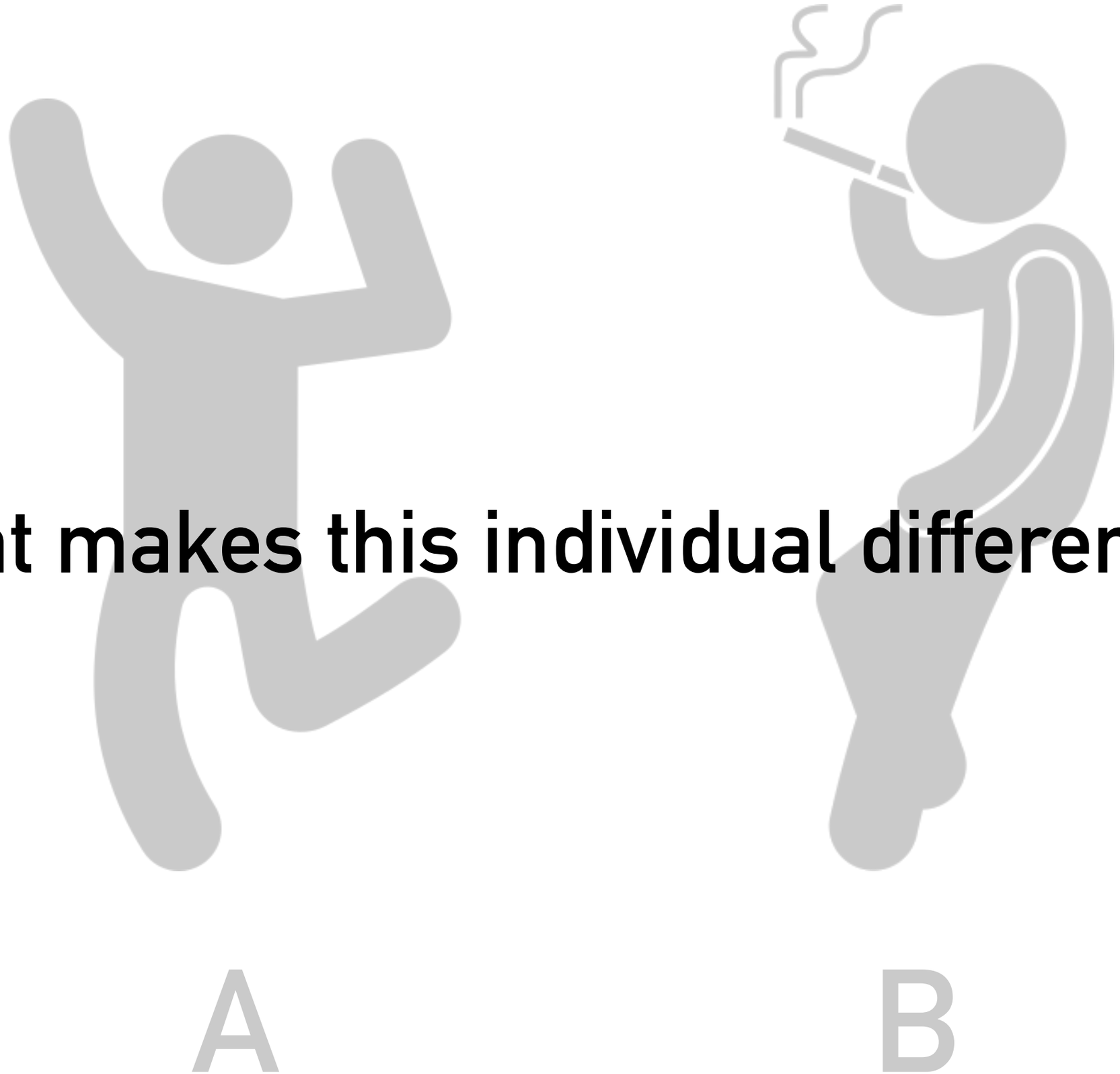
B

Tak nak!
EVERY BREATH YOU TAKE WILL EVENTUALLY SHORTEN YOUR LIFE.

60
50
40
30
20
10
0
how long can you live?
KOMITE NASIONAL
PENANGGULANGAN MASALAH MEROKOK
http://www.antirokok.or.id
Lintang
MAJALAH TREND PRIA

A

B



What makes this individual differences?

1. INTRODUCTION

- Relationship between behavior change and activity in subregions of the brain's medial prefrontal cortex (MPFC) during message exposure (Chua et al., 2011; Falk et al., 2011, 2012; Wang et al., 2013).

- Result of self-related processing? (Fishbein, 2011; Petty & Cacioppo, 1986)
  "Who think more about how a message relates to them might be those who go on to change their behavior"

- HOWEVER, explicit ratings of message self-relevance have not been found to mediate the MPFC-behavior relationship (Falk et al., 2010, 2011).

1. INTRODUCTION

1) Self-related processing does occur during message exposure, but its effects are not captured by previously used retrospective self-reports.

2) Different function, that is also executed within subregions of MPFC, is engaged during message exposure.

- Ventral MPFC plays a key role in representing the personal value of many types of stimuli during decision making (Grabenhorst et al., 2003; McClure et al., 2007; Rolls et al., 2003; Berns & Moore, 2012; Chib et al., 2009; etc)

- Value-related processing?
  "Who think the factor in message more valuable might be those who go on to change their behavior"

1. INTRODUCTION

- Hypothesis: Both are likely components of the persuasiveness of a message

- Independently identified self- and value- related regions, and a region of MPFC previously found to be predictive of behavior change.

- Examined the relationship of each of these three regions to smoking reduction in response to anti-smoking messages.

2. METHOD

- Participants (n = 50)
  - smoking at least 5 cigarettes per day for the past month
  - have been a smoker for at least 12 months
  - age of 18 ~ 65

- Study Timeline

Session 1

Intake appointment

- Baseline self-report smoking Questionnaires

Session 2

fMRI scanning appointment

- prescan and postscan self-report smoking Questionnaires
- fMRI scanning (self-localizer Task & Banner ads task)

Session 3

Follow-up appointment

- Smoking Questionnaires

1) Behavioral Data

- **Smoking Questionnaires**

  1) Intention Question (n = 3)
  - Intentions to quit, reduce, or refrain from smoking in the next 3 months

  2) Self-efficacy Question (n = 2)
  - How confident you are that you would be able to stop smoking in the next 3 months
  - How easy it would be for you to stop smoking

  3) Perceived risk Question (n = 5)
  - How much you thought that smoking can harm or had already harmed your health
  - How concerned you are that smoking had after your own health or someone else's health

  Behavior change: Session 3 (Follow-up) - Session 1 (Baseline)

- **Change in Daily Smoking**

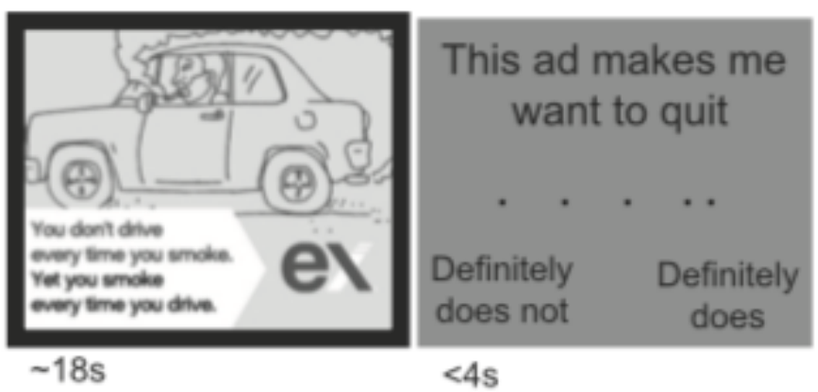
  Behavior change: (Session 3 (Follow-up) - Session 2 (Scanning)) / Session 2 (Scanning)

2) Imaging Data

- **Self-localizer Task**
  : Participants have to judge trait adjectives according to the conditions.

  ① You -> You
  ② You -> Friend
  ③ Friend -> You
  ④ Friend -> Friend
  ⑤ Valence

- **fMRI Banner Ads Task**
  : watch and rate 23 animated banner ads
  (mean length = 17.7 seconds)



2) Imaging Data

- **Self-localizer Task**

  : Participants judged the self-relevance and valence of trait adjectives.

① You -> You
② You -> Friend
③ Friend -> You
④ Friend -> Friend
⑤ Valence

Contrast Self-related judgements VS Valence

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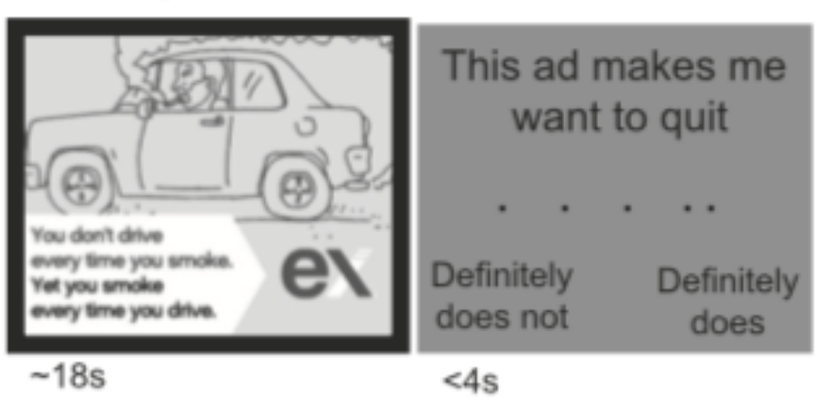
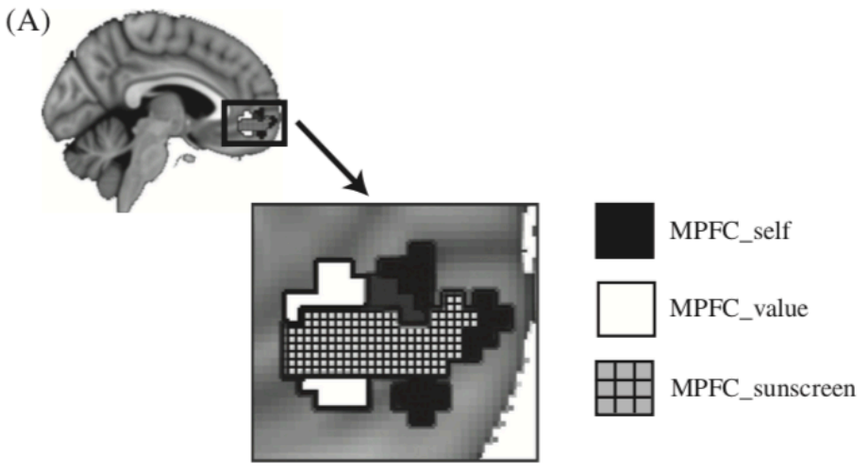


Contrast Neural activity during banner ads VS fixation rest periods

2) Imaging Data

- A Priori Regions of Interest



① self-related MPFC region

- brain region identified by the self-localizer task

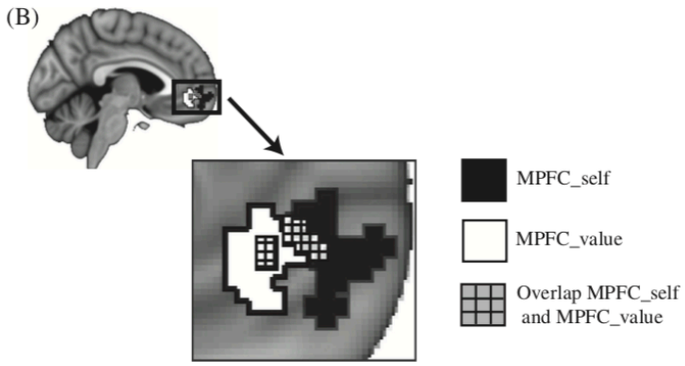
② value-related MPFC region

- taken from Barta et al., (2013)

③ MPFC region related to behavior change

- taken from a previous study that predicted changes in sunscreen use and smoking behavior change with neural activity

Additional Analysis



1. Overlapped regions

- MPFC_self_only voxels
- MPFC_value-only voxels
- Intersection voxels

2. Specificity of Predictive Effects

: Test whether the effects of interest were selective for the ads

① **Thinking about the self could predict behavior change?**

: Whether the ROIs' activity during self-related conditions (you -> you/ friend -> you) of self-localizer task could predict behavior change

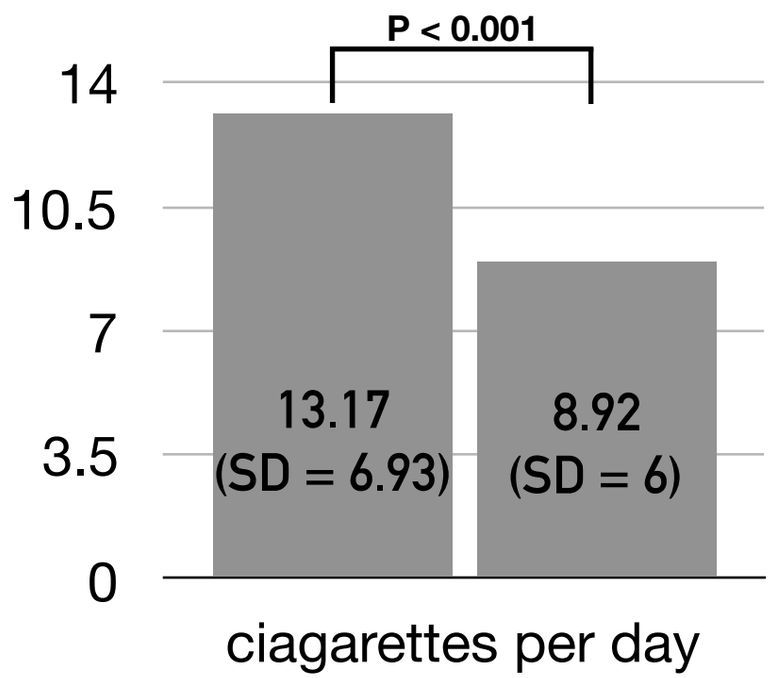
② **Mentalizing could predict behavior change?**

: Whether the ROIs' activity during "friend -> friend" conditions of self-localizer task could predict behavior change

3. Whole-brain analysis

3. RESULTS

- Smoking Behavior Change



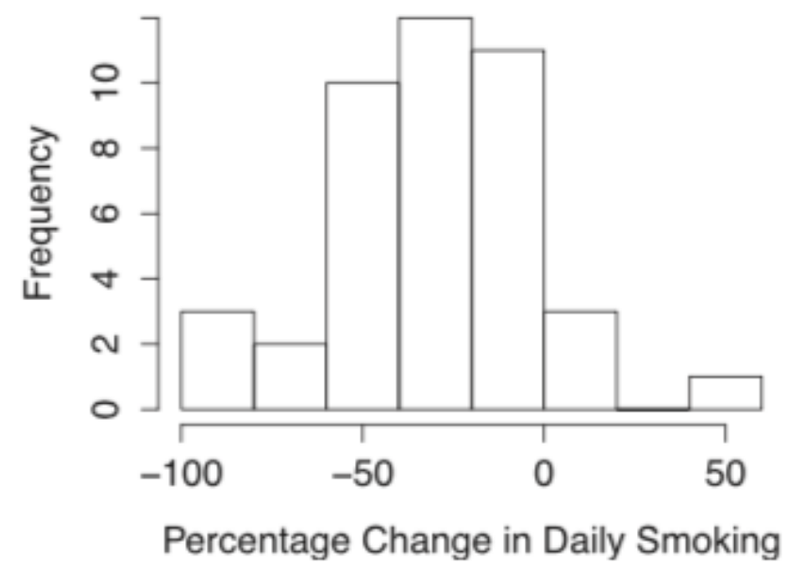


Table 1. Changes in smoking intentions, self-efficacy, and perceived risks

	Intentions*	Self-efficacy*	Perceived risks
Intake (Session 1)	2.42 (*SD* = 0.83)	2.65 (*SD* = 1.24)	3.27 (*SD* = 0.69)
Follow-up (Session 3)	3.08 (*SD* = 0.74)	3.95 (*SD* = 1.43)	3.25 (*SD* = 0.79)
Change (Session 3–1)	0.66 (*SD* = 0.78)	1.30 (*SD* = 1.37)	−0.02 (*SD* = 0.64)

Note. Measures that were significantly different between appointments are marked with an asterisk ($p < .005$).

3. RESULTS

- **Neural Activity During Ad Exposure Predicts Smoking Behavior Change**

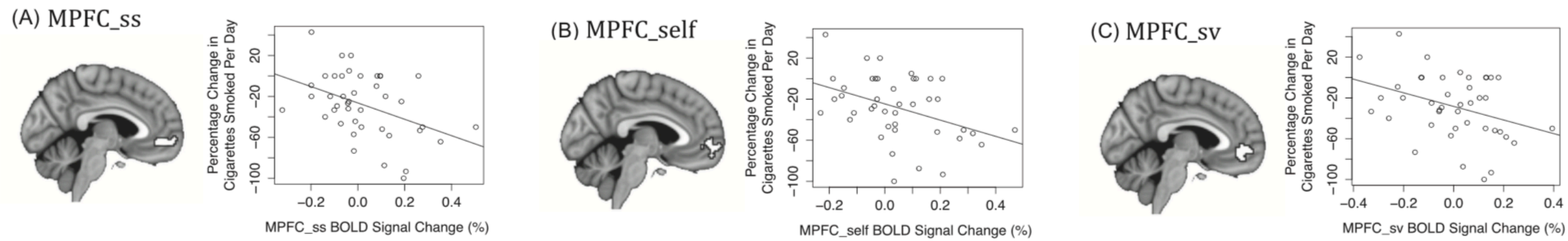


Table 2. Correlations between MPFC activity and changes in behavior and self-report metrics

		1	2	3	4	5	6	7	8
Behavior	1. Δ Smoking								
brain	2. MPFC_ss	−0.42*							
	3. MPFC_self	−0.40*	0.89**						
	4. MPFC_sv	−0.36*	0.87**	0.86**					
Self-report	5. Δ Intentions	−0.32*	0.00	0.14	0.06				
	6. Δ Self-efficacy	−0.19	0.05	−0.04	0.01	0.37*			
	7. Δ Perceived risk	−0.27	0.18	0.07	0.17	0.34*	0.36*		
	8. Ad quit rating	−0.16	−0.03	0.05	−0.08	0.02	0.10	−0.09	

Notes. MPFC = medial prefrontal cortex. *$p < .05$; **$p < .005$.

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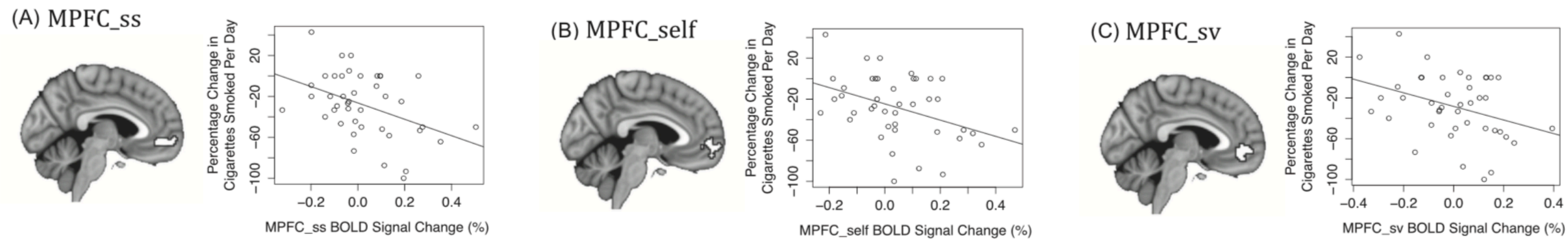

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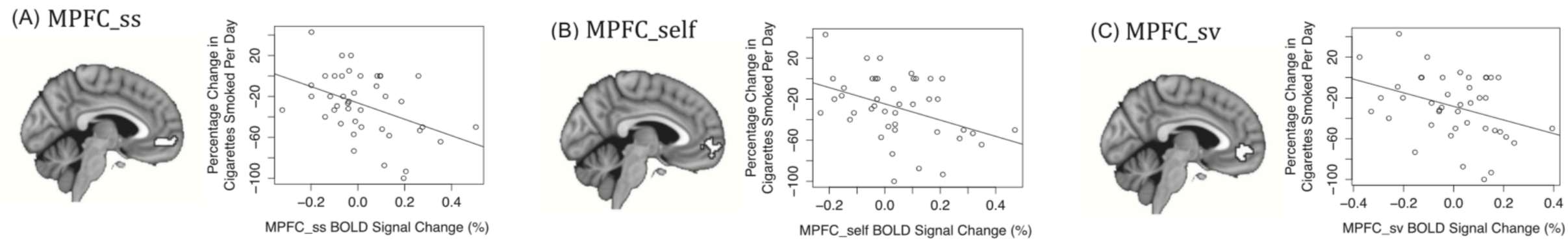

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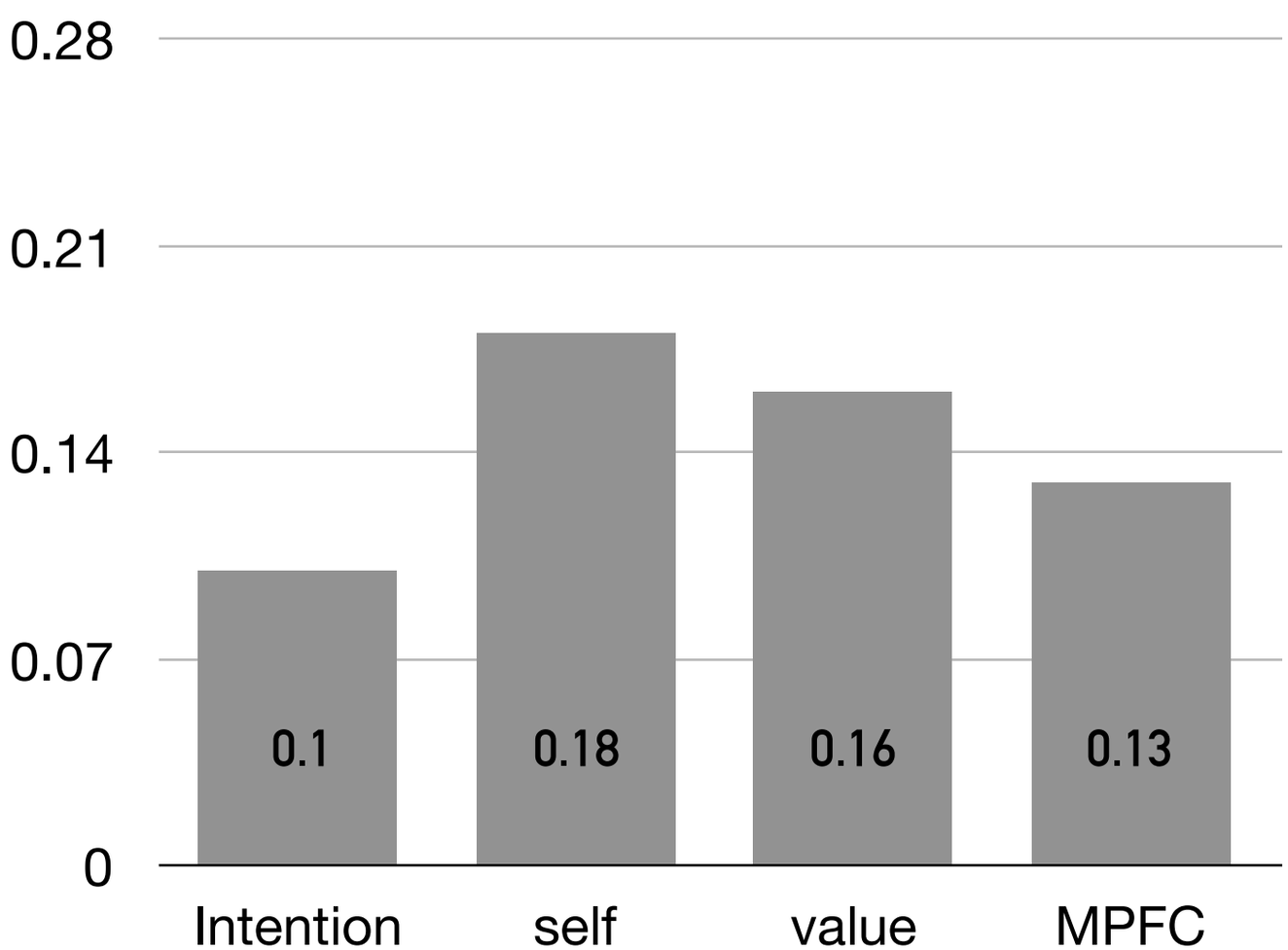
3. RESULTS

- R^2 of each measure



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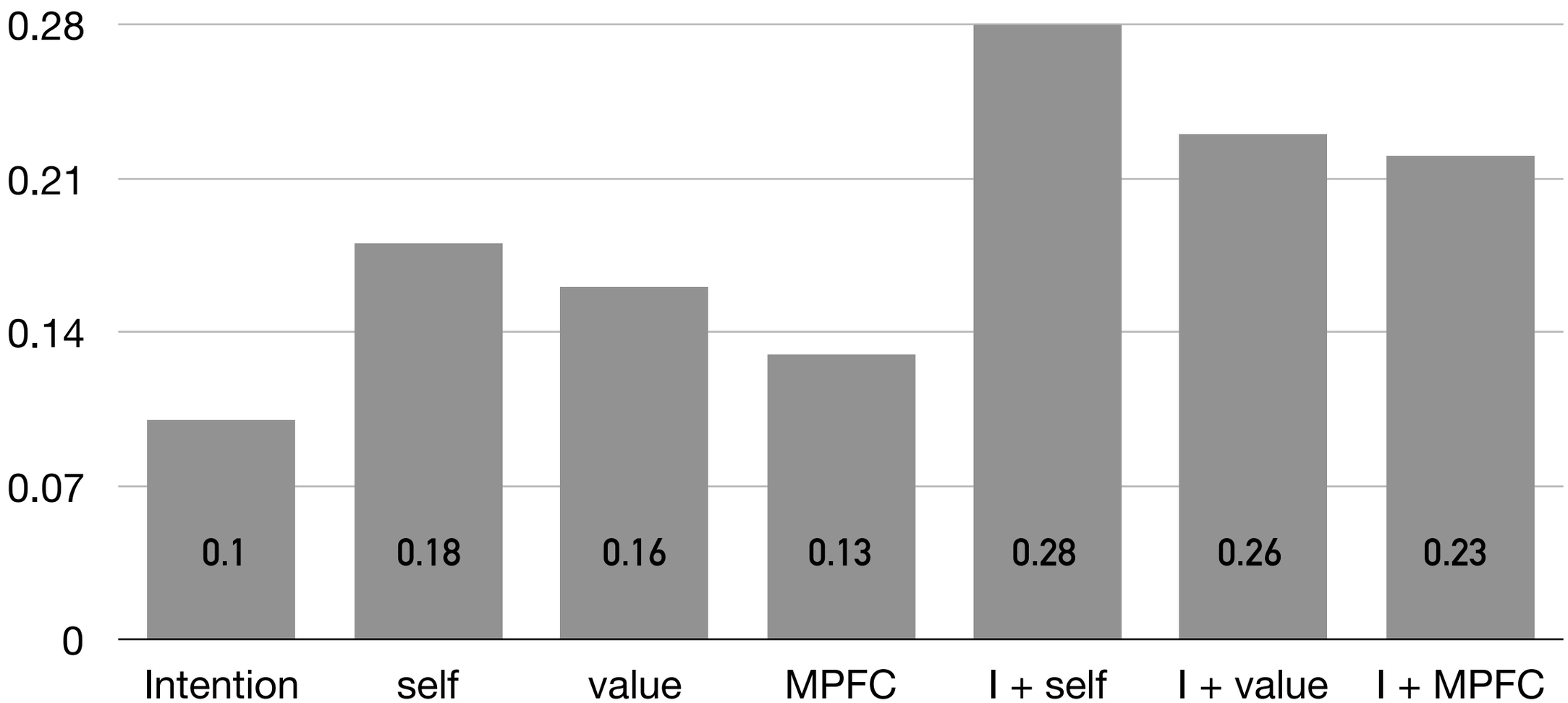


3. RESULTS

1. Overlapped regions, self- only regions, and value- only region all predicted reductions in smoking

	Beta	P-value
MPFC_self_only*	-0.8	0.008
MPFC_value_only*	-0.66	0.02
Intersection	-0.62	0.014

2. Specificity of Predictive Effects

① None of the ROIs during self task condition predicted behavior.

② None of the ROIs during mentalizing condition predicted behavior.

3. Whole-brain analysis

Table 3. Whole brain exploratory analysis of correlations with behavior change

Region	Lat.	Size	Peak T	X, Y, Z
Medial prefrontal cortex	L (/R)	92	−3.54	−16, 56, −8
Temporoparietal junction*	L	46	−3.35	−44, −40, 28
Medial temporal lobe*	L	31	−4.59	−13, −5, −38
Parahippocampal gyrus*			−2.42	−18, −5, −36
Parahippocampal gyrus*	L	47	−3.89	−20, −26, −14
Lateral occipital cortex	R	25	−3.63	35, −84, 1
Cerebellum	L	27	−4.01	−33, −30, −35

Notes. Negative activations were associated with reduced smoking. No regions showed a positive effect in this contrast (i.e., no regions were associated with increased smoking). *$p < .005$, $k = 20$.

4. DISCUSSIONS

- Replicated that self-related processing engage in MPFC - behavior relationship, by using functional localization approach.

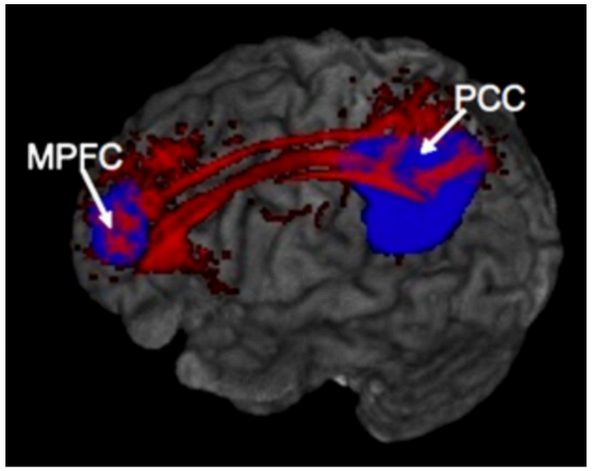
- Suggests that value-related processing might also play a role in explaining the relationship.

- Hippocampus has been linked with memory, prospection, and imagining in the future (Buckner & Carroll, 2007; Schacter, Addis, & Buckner, 2007; Schacter et al., 2012; Spreng, Mar, & Kim, 2009). Therefore, people who can do more imagination while watching the advertisement might be better at smoking behavior change.

4. DISCUSSIONS

< Limitations & Future directions >

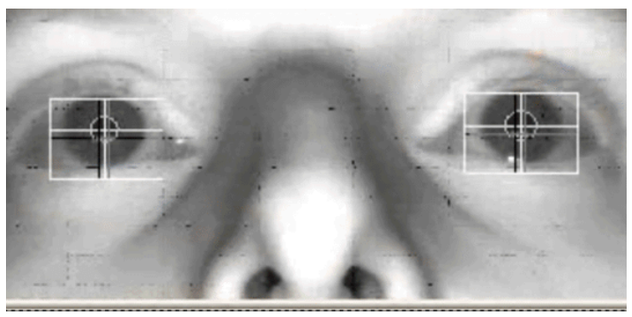
- The result cannot establish causal media effects on subsequent behavior
- Functional localization for value-related processing can be used.
- Disentangling of self- and value- related processing.
- Brain network analysis - Default Mode Network
- Effects of multiple exposure of the messages.



The default mode network includes the medial prefrontal cortex (MPFC) and the posterior cingulate cortex (PCC).

5. Research plan

Session 1
Follow up session
Daily smoking
Intention to Quit
0:04 / 0:42
0:27 / 0:42
Intention to Quit
Daily smoking
Intention to Quit



Contempt
Sadness
Surprise
Happiness
Neutral
Anger
Fear
Disgust

Hypothesis

1. Less attentional bias toward smoking cues and less neglect of smoking's side effects would predict behavior change.
   - The more and longer eye fixation when smoking cue appears compared to the depiction of side effects of smoking, the less behavior change.

2. Positive facial expression toward the smoking cues and negative facial expression toward the side effects would negatively correlate with behavior change.

3. Replicate the findings of MPFC activity - behavior change relationship with fNIRS.

4. Combination of eye movement, facial expression, and neural data would have better prediction compared to neural data alone.

THANK YOU